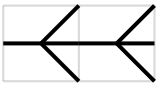
Trøndelagkonkurranse 2026 - Level II

Second Round - November 3, 2026

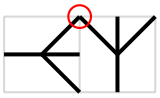
2nd Part

Nils draws an Algiz rune inside a square

Then Bianca asked him, "Can you arrange 16 of those on a 4-by-4 grid such that the runes do not touch each other in the corners of the squares?"



Allowed



Not allowed

Task 1: Help Nils figure out if that is possible or not.

Task 2: Now, help Nils figure out if the same is possible for an n-by-n grid where $n \geq 2$.

Nils is very grateful that Bianca gave him such a nice problem, so he would also like to give Bianca a similar problem, either using grids that are not necessarily rectangular, runes with different shapes, or anything interesting he can think of.

Task 3: Help Nils create a nice problem he can give to Bianca.

Task 1: It is NOT possible. Notice that a 4-by-4 grid will have to contain 16 runes and each rune covers 2 vertices, so 16 runes must cover a total of 32 vertices. However, a 4-by-4 grid has 25 vertices, so it is not possible to arrange the runes without some runes touching in the corners.

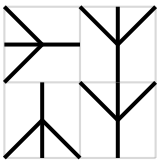
Task 2: In general, let's observe that an n-by-n grid will require n^2 runes so they will cover $2n^2$ vertices. On the other hand, an n-by-n grid has $(n+1)^2$ vertices, so the arrangement of runes is possible as long as

$$2n^2 \leq (n+1)^2,$$

which is equivalent to

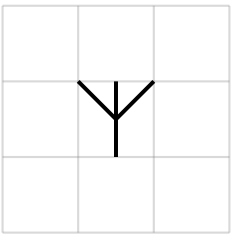
$$(n-1)^2 \leq 2,$$

and this is only satisfied for $n = 2$. Here is one way to do it:

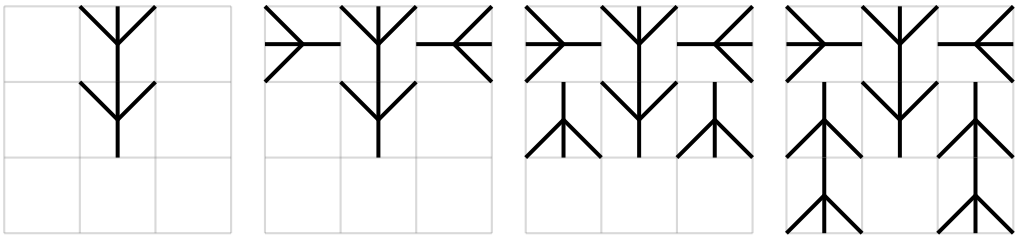


Alternative approach:

Any n-by-n grid of $n \geq 3$ must contain a 3-by-3 grid. Since a 3-by-3 grid has rotational symmetry, let's start with an Algiz rune in the middle point upwards as follows:



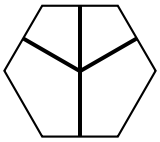
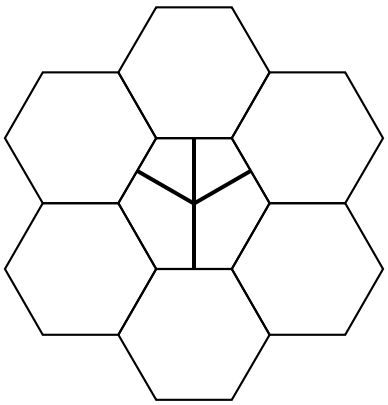
Then the top cell can only have a rune in the same position (a); then the top left and top right cells can only have a rune in one position each (b); then the middle left and middle right can only be in one position each (c); then the bottom left and bottom right can only be in one position each (d). Finally, the middle bottom cell has already all its vertices covered, so no rune can be placed here without touching other runes. In conclusion, it is not possible to arrange runes as desired for n-by-n grid with $n \geq 3$.



(a) (b) (c) (d)

Task 3: Johannes Brendberg from Guri Kunna Vgs proposed the following problem:

Consider an Algiz rune inside a regular hexagon as follows:



Can 7 of these be arranged on a 3-honeycomb (below) such that no two ends of a rune meet on the edges of the 3-honeycomb? (This includes the four ends of the runes)